

JOHANS UTAMA

BACHELOR OF ENGINEERING: ARTIFICIAL INTELLIGENCE

+86 19705199251

 johansutama25@gmail.com

 Johans Utama

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SUMMARY

AI engineering student and sole-author researcher, recognized with multiple academic scholarships for outstanding performance. Builds and evaluates systems across computer vision, information retrieval, and NLP, with several algorithms implemented from scratch. Multilingual (Indonesian, English, Chinese) with experience in cross-cultural technical collaboration.

EDUCATION

Bachelor of Engineering: Artificial Intelligence

Sep 2023 - Jun 2027

Nanjing University of Information Science and Technology | Cumulative GPA: 4.19 / 5.00

EXPERIENCES

- Indonesia NUIST Association

Nov 2023 - Jun 2024

 - Served as internal secretary
 - Streamlined internal communications, maintained official documentation, and coordinated activities among board members
- Indonesian Students Association in China, Nanjing Branch

Feb 2024 - July 2025

 - Served as an active member of the Education Department for 2 consecutive periods
 - Executive Secretary for the 2024 Chaanakya Chinese Language Competition
 - Head of Creative Business Division for the 2025 Nanjing Nusantara Festival
 - Head of Financial and Sponsorship Division for the Indonesian Students Leadership Camp
- Academic Affairs Department

Sep 2024 - Jun 2025

 - Organize weekly on-campus events centered around language and cultural exchange
- Indonesian Students Association in the People's Republic of China

Oct 2024 - July 2025

 - Member of Arts, Culture, and Tourism Department
 - Overseeing the Indonesian Culture Content Program
 - External Relations Officer of the Indonesian 2025 National Congress

AWARDS & CERTIFICATIONS

Awards	Certifications
- International Climate Science Olympiad Semifinalist - Jiangsu Government Excellent International Student Scholarship (2024-2026) - NUIST First-class Scholarship (2024-2027) - NUIST Excellent Freshmen Scholarship (2023)	- Search Engine Optimization - Generative AI - Natural Language Processing on Google Cloud - Introduction to Artificial Intelligence - Introduction to Machine Learning

SKILLS

Natural Language Processing	Python	Machine Learning
Computer Vision	Deep Learning	Microsoft Office

LANGUAGE

- Indonesian (Native)
- English (Proficient, IELTS 7.0)
- Chinese (Intermediate, HSK 4)

PROJECTS

T5-Based Formal-Informal Text Style Transfer System

Oct 2025

- Fine-tuned T5-base for formal-informal text rewriting
- Built merged dataset of 12.7K parallel sentence pairs
- Trained under low-resource GPU constraints (8GB VRAM)
- Evaluated with EMR, Jaccard, content preservation, and style scoring
- Analyzed performance limitations under small-data constraints

Urban Waste Detection System

Nov 2025

- Built a YOLOv8s-based object detection model on a 3K image dataset for automated trash detection.
- Applied COCO transfer learning and advanced augmentation (Mosaic, MixUp) to improve generalization in cluttered urban scenes.
- Achieved 0.75 precision and 0.53 mAP@0.5 with ~50ms GPU inference (near real-time performance).
- Performed precision-recall trade-off analysis and identified strategies to improve small-object detection and recall.

Document Retrieval Engine with TF-IDF and BM25 Ranking Analysis

Dec 2025

- Implemented a document retrieval system using TF-IDF and BM25 probabilistic ranking
- Evaluated using Precision@5, AP, MAP
- Demonstrated query-dependent trade-offs between vector and probabilistic models
- Designed modular indexing and query processing architecture

PUBLICATIONS

Utama, J. (2026). "Algorithmic Bias in Social Media Recommendation Systems: A Systematic Review of Political Filter Bubble Evidence (2019–2026)." *SSRN Preprint*. [\[SSRN\]](#)

Aug 2026

Utama, J. (2026). "Deep Learning-Based Meteorological Data Downscaling: A Comparative Study with Physics Informed CNN and a Component-Level Ablation Analysis." *EarthArXiv Preprint*. [\[EarthArXiv\]](#)

Aug 2026